

Adversarial Auto-encoders for Image Generation from standard EEG features

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Abstract—New deep-learning-based classification and generative models have recently found applications in EEG and fMRI-based brain studies. In this manuscript, we propose a simple solution of finding feature representations of EEG signals using statistical analysis which can be used for classification of stimuli in real time. Also, we use a deep generative model called adversarial auto-encoder (AAE) and attempt to reconstruct the stimuli back from these features. The EEG data was recorded from visual stimuli of subjects who were shown images from the ImageNet dataset and the proposed model aims to reconstruct images directly from the EEG feature manifold. Classification accuracy of 90% was achieved for the classification of images based on EEG features. Qualitative evaluation of the reconstructed images demonstrates the efficacy of our experimental model. The models can be trained on a PC with classification in real time

Keywords— EEG; GAN; Deep Learning; adversarial auto-encoder; Image Generation

I. INTRODUCTION

Electroencephalography (EEG) is the study of electric impulses produced by the brain. The signals that are generated by the brain are the direct indicator of what the person might be thinking or is trying to do with his/her own body. The study of the EEG signals is useful in many research areas and has various applications such as detection and classification of event related stimuli, seizure detection [1] and prediction, study of mental disorders like psychiatric disorders and dementia, brain-computer interfacing and sleep/dream analysis [2, 3]. The EEG signals are further divided into five major sub-bands based on the frequency ranges for better understanding of human behavior. These bands from low to high frequencies are called delta (Range 0.5-4Hz), theta (Range 4- 8Hz), alpha (Range 8-13Hz), beta (Range 13-30Hz) and gamma (Range 13-30Hz) frequencies.

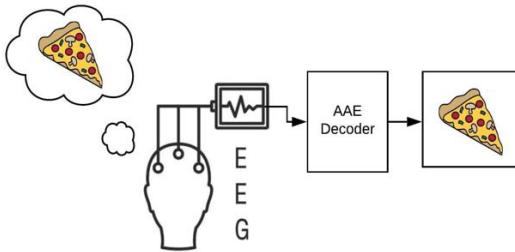


Fig. 1. A basic structure of the proposed model for image reconstruction from EEG.

Recent advancements in deep learning techniques, such as generative adversarial networks (GANs) and auto-encoders have opened up many application possibilities for these networks [4,5]. While auto-encoders mostly find applications in image compression, denoising etc. [13], GANs have found their applications in image and sequence synthesis, drawing artworks [6], 3D models, and so on. Adversarial auto-encoders (AAEs) are a very recent invention and demand more research work in order to use them in broader applications. AAEs have a very high potential especially in use-cases such as reconstructing images from signal features. Even though there are few research articles published on generating images using EEGs without using AAE, the results are not very practically accurate. Hence, here is scope for improvement in this area and also a possibility of a breakthrough in the understanding of the enigma, which is our brain. In this manuscript, we propose to build a conditioned AAE model, which takes in features from brain signals (preferably functional magnetic resonance imaging (fMRI) and EEG etc.) after preprocessing and be able to generate accurate enough approximations of what the person is thinking. This research could help in not just improving the understanding of how the brain functions when it uses the visual cortex but also use its capabilities to generate images on-the-go. This unlocks new possibilities in the field of Neuro- science. This could also find its application in dream analysis, psychotherapy and also understanding and possibly help people without vision or with neurological conditions.

In the proposed work, we have generated images from EEG signals. The proposed work provides an improvement over the experiments done by Kavasidis, I. et. al., to produce images from brain signals [7]. The data set for the work has been taken from [7]. The authors in [7] have used long short-term memory (LSTM) networks as a discriminative model and then in the later stages compared GAN and variational auto-encoder (VAE) as their generative models. The proposed work replicates the results of previous papers [7,8] in a framework that is very easy to train and works in real-time with very low computational power requirements. To achieve that, we first decomposed the EEG sequences into wavelets using Daubechies series and then features were extracted from these wavelets using statistical techniques. It was important to know if the features extracted using the statistical approach could be distinguished by the machine learning models into different classes. So, the feature vectors were passed through an artificial neural network (ANN) classifier and it was nearly 90% accurate in distinguishing the features.

VAEs and GANs are both very popular generative models, but they have their own drawbacks like overfitting in VAE and high training instability in GANs [5]. In order to overcome

these disadvantages of VAE and GAN, the employment of AAE as a generative model is proposed as it combines the forgery technique used by GAN and the semi supervised learning technique of VAE to produce a model which has the advantages of both but the disadvantages of none.

A. Related Work

Brain signals have proved to be useful in BCI and medical anomaly detection applications. There are several techniques such as EEG, fMRI etc. which can be used for recording brain activities. EEG is a low-cost technique which can provide higher temporal resolution than fMRI and it proves to be more advantageous when compared to other neuroimaging techniques. Even though fMRI has huge potential applications, it fails to be as cost effective as EEG. Image generation using feature vector is a hot topic in research right now. Several techniques like VAEs, GANs, AAEs etc. have been developed for generating images from feature vectors. AAE seems to outperform the other two techniques as it combines the advantages of GAN and VAE [5]. Spampinato, C. et al. have used EEG sequences and encodes them into features using an LSTM architecture. Further, they have applied convolutional neural networks (CNNs) to do regression from images to the feature vector showing that classification of EEG is possible [8]. As a continuation, Kavasidis, I. et. al. used GANs and VAE for image reconstruction. Later, the concept of manual feature picking instead of using LSTM architecture is proposed for computational simplicity [2]. The proposed work utilizes the concept of analyzing statistical features instead of LSTM sequence analysis. This has resulted in lesser computational complexity, while achieving almost equivalent results as in [8]. The concept of AAE and its utilization for image construction has been described in [4,5]. The performance of AAEs for many tasks has been more accurate than GANs. The proposed work aims to reconstruct images from EEG, and the superiority of AAEs over GANs is demonstrated by experimental results. The manual features were chosen to be statistical measures like kurtosis, skewness, mean, max, and standard deviation as suggested in [2].

Since we chose a small subset of the dataset for training the AAE because of limited computational power we decided to modify it by multiplying the EEG features with Gaussian distribution to make the training easier which was inspired by the proposal in [9]. Discrete wavelet transform (DWT) is being used for feature extraction as it provides both time and frequency domain features extraction and is suitable for non-stationary signals like EEG. The proposed work provides an enhancement over the existing works for generating images from EEGs as it uses handpicked statistical features resulting in decreased computational complexity. It also utilizes a modified AAE architecture that overcomes the existing problems of GANs and VAEs.

II. MATERIALS & METHODS

A. Dataset

We have used the publicly available dataset for this research work which is provided by perceive labs [7]. To prepare this dataset, six subjects were chosen to have similar age, education level, and cultural background. They were shown images of various categories present in IMAGENET while the EEG data was being recorded. The images were shown at a burst of 0.5s per image. The characteristics of the

dataset are summarized in Table I. 4 out of the 40 classes in the dataset were randomly chosen to test our hypothesis.

TABLE I. CHARACTERISTICS OF THE DATASET.

Number of classes	40
Images per class	50
Number of EEG channels used	128
Sampling frequency	1000Hz
Data resolution	16bit

A 128 channel device was used to collect the EEG signals. A channel wise plot of the EEG signals with amplitude on Y-axis and time on X-axis has been shown in the figure 2. A time sample of 0.5s was taken.

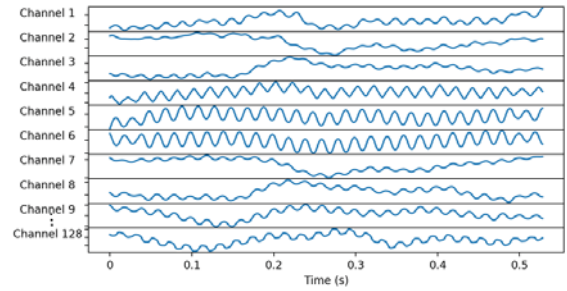


Fig. 2. Channel-wise EEG amplitudes plotted for a random subject.

B. Feature Extraction

As the aim is to translate brain activity into images, we would first need to generate/acquire a dataset specific to the task. We use the EEG mentioned in the previous section to extract features. We applied a classical feature extraction technique for signals such as DWT that encodes the EEG into a feature vector as done in [2]. Extracted wavelet coefficients provided both time and frequency representation of the EEG samples. Several pilot experiments were carried out with different types of wavelets and db13 turned out to be the best in terms of classification accuracy. We have applied Daubechies-13, level-4 wavelet decomposition to acquire the coefficients. A4, D4 and D3 coefficients were chosen, and then we used those coefficients to calculate the features as given in the Table II.

Most of the features mentioned above are common statistical features except kurtosis and skewness. Kurtosis is the measure of narrowness or the flatness of the signal. High kurtosis indicates a sharp peak at the mean whereas low kurtosis indicates a flat peak at the mean. Skewness is a measure of asymmetry in the signal with its value equal to zero for symmetric signal and less than or more than zero depending on the longer tail being on the left or right side respectively.

TABLE II. DIFFERENT STATISTICAL FEATURES EXTRACTED FROM A4, D4 AND D3 COEFFICIENTS OF DWT OF EEG SIGNALS

Name of the extracted feature	Mathematical description
Mean	$\mu = \frac{\sum_{i=2}^n x_i}{n}$
Kurtosis	$K = n \frac{\sum_{i=2}^n (x_i - \mu)^4}{(\sum_{i=2}^n (x_i - \mu)^2)^2}$

Standard Deviation	$\sigma = \left(\frac{\sum_{i=2}^n (X_i - \mu)^2}{n} \right)^{1/2}$
Maximum	$M = \max_{i=1}^n X_i$
Skewness	$S = \sqrt{n} \frac{\sum_{i=2}^n (X_i - \mu)^3}{(\sum_{i=2}^n (X_i - \mu)^2)^{3/2}}$

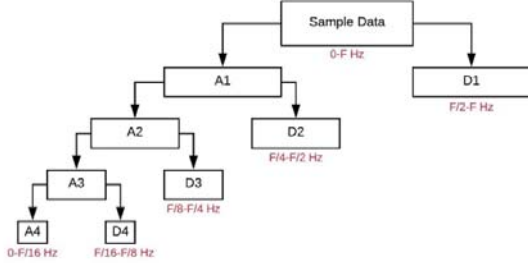


Fig. 3. Discrete wavelet decomposition up to four levels. A4, D4 and D3 wavelet features are chosen for classification step.

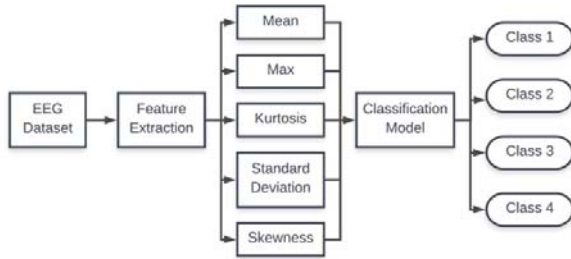


Fig. 4. Flowchart of EEG classification process. As shown, the five features were used as input for different classification models and were classified into the four randomly chosen classes (pizza, pool table, missile, parachute) out of the 40 available in the dataset.

These features were calculated for all the 128 channels and concatenated into the final extracted features file. So, for each EEG reading, we get $5 \times 3 \times 128 = 1920$ features. As the dimension of feature vectors was high, we applied principal component analysis (PCA) to reduce the dimension. For real-time applications, the same technique can be incorporated on a windowed EEG signal for feature extraction and then classification.

We selected 40 dimensions constituting 95% of the variance from the original feature set. These features were used to start the classification process. The feature set is then further split into 85% train set and 15% test set (later used for validation) for the classification step.

C. EEG Classification

For classification of EEG, we experimented with both classical machine learning (ML) models like support vector machine (SVM), and random forest classifiers as well as a deep learning model [10]. Classification is done on four output classes from ImageNet (n03888257 = parachute, n07873807 = pizza, n03982430 = pool table, n03773504 = Missile). The classes were randomly selected for the purpose of demonstration of classification accuracy. The parameters for the techniques which we have applied for the classification task of EEGs are mentioned briefly below:

1) SVM

We used the Gamma radial basis function (RBF) kernel function for training the features in SVM. The criteria chosen for this as well as the other techniques were found to be the most accurate through trial and error [11,12].

2) Random Forest

The random forest model which we used has 30 numbers of estimators (i.e. the number of trees in the forest model), also Gini impurity criterion was used.

3) Deep Learning Model

The ANN model architecture which we used is described in Table III.

TABLE III. ANN MODEL ARCHITECTURE

LAYERS	NODES
Input	40
Dense(ReLU)	128
BatchNorm	-
Dropout	-
Dense(ReLU)	64
BatchNorm	-
Dropout	-
Dense(ReLU)	32
BatchNorm	-
Dropout	-
Output	4

Activation function: SoftMax, Loss Function: Categorical cross entropy, Optimizer: Adam

We obtained a classification accuracy of nearly 90%. The number of classes chosen had to be limited because of the complexity involved with the next step of image generation and lack of enough resources for training thousands of images through the AAE

D. Image Generation

Getting an accuracy of above 90% validates the hypothesis that the acquired EEG signals reflect the images shown to the users. Hence, a generative model may be applied to generate the images. We moved forward to the step of generating image just from the EEG feature vector. Recently, deep learning-based generative models have gained a lot of attraction. GANs and VAEs are most commonly used methods for image generation. A traditional GAN consists of two neural networks, one called the generator and the other called discriminator. They play a game of cat and mouse and in turn help each other with model accuracy. Discriminator network is a simple classifier that classifies an input image into whether it is fake or real, i.e. if it's generated by the generator network or if it's part of the dataset. The generator is a de-convolutional neural network that takes in random noise and up samples the input into an image through a process called deconvolution which in simple terms is just the opposite of a convent that classifies input images into labels. Both nets are adversaries to each other (hence the name) and they continuously optimize each other due to opposing loss functions by playing a min-max game. Their losses push against each other and therefore it's theoretically possible to obtain a generator so good that the discriminator cannot predict if the output is fake, even if it is a slightly different version of the data. The overall objective

function of GAN can be written as in Equation (1) shown by [4].

$$\max_D \min_G V_D(D, G) = \max_D \min_G \left(E_{x \sim p_{data}} [\log D(x)] + E_{z \sim p_z} [\log (1 - D(G(z)))] \right) \quad (1)$$

However, a new generative model called AAE [5] was recently proposed, which performed inference by matching the posterior of the latent vector of the auto-encoder with a prior distribution.

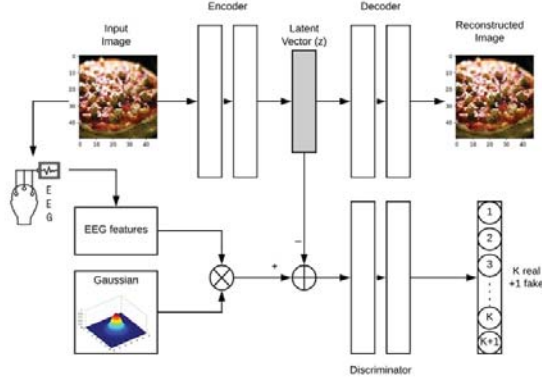


Fig. 5. AAE with EEG weighted Gaussian distribution as prior process.

AAEs use an auto-encoder structure coupled with an additional adversarial discriminator unit. The hidden vector (also called the latent vector) of the can be matched to any arbitrary distribution (denoted as) by using adversarial loss from a simple discriminator network. During training, the auto-encoder tries to reconstruct the given input data while the adversarial unit (discriminator) tries to predict whether the sample arises from the hidden vector of the auto- encoder or from a user-specified vector.

The aggregated posterior distribution (denoted as) of the latent/hidden vector of the auto-encoder can be written as in Equation (2) shown by [5].

$$q(z) = \int_x q(z|x) p_d(x) dx \quad (2)$$

Here, $q(z|x)$, $p_d(x)$ are the encoding distribution and the data distribution respectively.

AAE is similar to VAE because they both match the distribution of the latent vector. But, VAE uses Kullback-Leibler (KL) divergence, and AAE uses adversarial training. The main difference between the two is that for VAE we need to have access to the exact function of the prior distribution, however in AAE we only need to sample from prior distribution which makes it possible to map highly complex distribution. Additionally, GANs impose the data distribution at a pixel level on the output of the network.

However, AAE relies on the training to mimic the data distribution, which results in better test-likelihood. Because of the aforementioned reasons we decided to proceed with AAE for our experiments. For the problem of image generation, directly from EEG signals, we had to make two modifications to the traditional AAE structure to satisfy our goal. Firstly, we added a multi-class output layer in the discriminator instead of a traditional binary (real/fake) label.

We used a K+1 sized one- hot vector for K real and 1 fake class. This proved to perform better than a simple discriminator for our multi-class problem. In addition to that, we weighted the Gaussian distribution with the EEG feature set before using it as the prior distribution for the hidden vector of the AAE. The prior distribution is mathematically formulated as in Equation (3).

$$p(z) = X_{EEG}(i) * z, \forall z \sim N(0,1) \quad (3)$$

Where, $X_{EEG}(i)$ is the EEG feature vector measured for the image and z is a Gaussian with zero mean and unit standard deviation. The size of the image passed into the encoder of the auto- encoder was kept at (50 50) which we found to be the right trade-off for computational complexity and image clarity. Two Dense layers of 1000 nodes, each with Leaky ReLU activation function were used for both the encoder and the decoder net- work. The Discriminator network has two dense layers with decreasing node size and an output layer with SoftMax activation for the K+1 possible classes. The network was trained on categorical cross-entropy loss function.

The latent vector has 40 dimensions as we have a weighted prior with EEG feature vector size of 40 after applying PCA. Our AAE model was trained on the 4 test ImageNet class images first with unit EEG values to avoid overfitting. The overall structure of our proposed model is shown in Figure 5.

III. RESULTS & DISCUSSION

A. EEG Classification

Training accuracy as well as Test accuracy obtained by each classifier model discussed in Section 3.3 is shown in Table IV. As stated earlier, the dataset was divided into 85% train set and 15% test set.

TABLE IV. TESTING AND VALIDATION ACCURACIES

MODEL	Training Accuracy	Test Accuracy
SVM	-	76.5
Random Forest	-	80.3
ANN	91.2	89.8

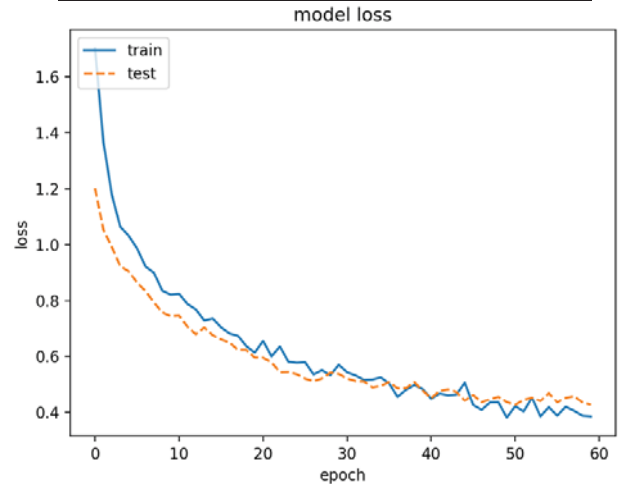


Fig. 6. ANN loss curve over 60 epochs

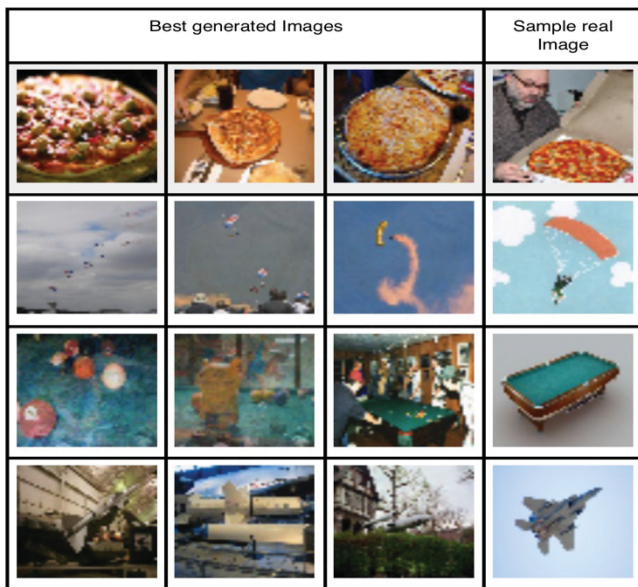


Fig. 7 The best construction result of our AAE MO

The high classification accuracy on the EEG features validate our hypothesis that handpicked features for EEG from visual stimuli can be classified and thus we can make use of these features for image generation instead of training a computationally expensive model like LSTM to learn these features.

IV. CONCLUSION

In this experiment, we validated the hypothesis of classification of visual stimuli-based EEG signals just based on handpicked statistical features and tried to generate image re-construction from EEG samples using a modified adversarial network. Our work proved that proper feature selection for this EEG classification task can result in a high accuracy which we tested using both traditional ML models and deep learning. Also, satisfactory results for the image reconstruction from EEG using AAE with very limited use of computational power as well as a small dataset proved that this research could have achieved a much better reconstruction accuracy with a larger as well as deeper convolution layer models. In future, we would like to go deeper into classification of even wider variety of EEG stimuli data by

possibly clustering the latent vector of the AAEs. We will be modifying our current model such that it generates video frames instead of just images. Once we are successful, we would like to experiment our new model on dream analysis.

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